FS5VS-10

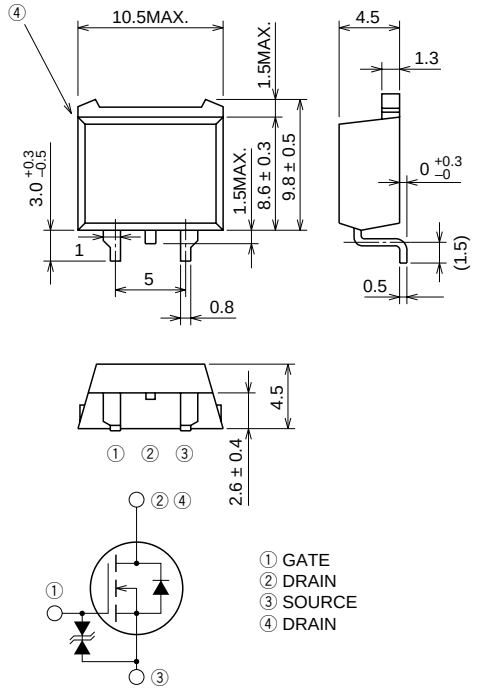
HIGH-SPEED SWITCHING USE

FS5VS-10



- V_{DSS} 500V
- $r_{DS\ (ON)\ (MAX)}$ 1.8Ω
- I_D 5A

OUTLINE DRAWING Dimensions in mm



TO-220S

APPLICATION

SMPS, DC-DC Converter, battery charger, power supply of printer, copier, HDD, FDD, TV, VCR, personal computer etc.

MAXIMUM RATINGS (T_c = 25°C)

Symbol	Parameter	Conditions	Ratings	Unit
V_{DSS}	Drain-source voltage	V_{GS} = 0V	500	V
V_{GSS}	Gate-source voltage	V_{DS} = 0V	±30	V
I_D	Drain current		5	A
I_{DM}	Drain current (Pulsed)		15	A
P_D	Maximum power dissipation		90	W
T_{ch}	Channel temperature		–55 ~ +150	°C
T_{stg}	Storage temperature		–55 ~ +150	°C
—	Weight	Typical value	1.2	g

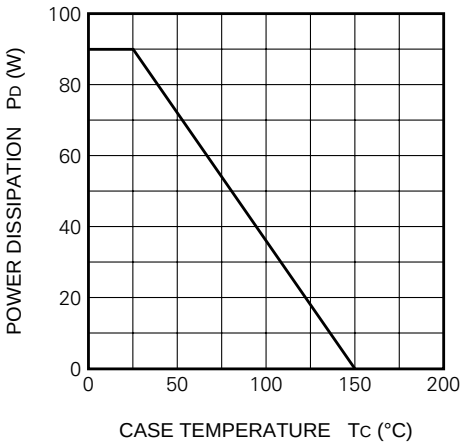
MITSUBISHI ELECTRIC

ELECTRICAL CHARACTERISTICS (Tch = 25°C)

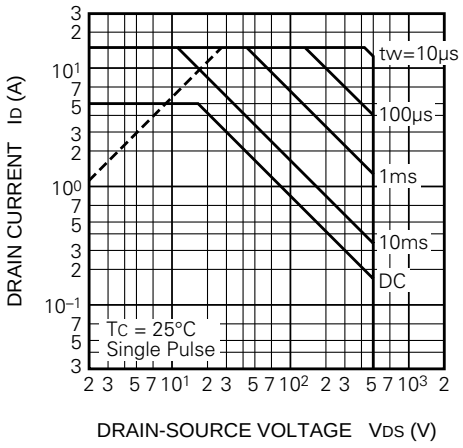
Symbol	Parameter	Test conditions	Limits			Unit
			Min.	Typ.	Max.	
$V_{(BR)\ DSS}$	Drain-source breakdown voltage	I_D = 1mA, V_{GS} = 0V	500	—	—	V
$V_{(BR)\ GSS}$	Gate-source breakdown voltage	I_G = ±100μA, V_{DS} = 0V	±30	—	—	V
I_{GSS}	Gate-source leakage current	V_{GS} = ±25V, V_{DS} = 0V	—	—	±10	μA
I_{DSS}	Drain-source leakage current	V_{DS} = 500V, V_{GS} = 0V	—	—	1	mA
$V_{GS\ (th)}$	Gate-source threshold voltage	I_D = 1mA, V_{DS} = 10V	2	3	4	V
$r_{DS\ (ON)}$	Drain-source on-state resistance	I_D = 2A, V_{GS} = 10V	—	1.4	1.8	Ω
$V_{DS\ (ON)}$	Drain-source on-state voltage	I_D = 2A, V_{GS} = 10V	—	2.8	3.6	V
$\lvert y_{fs} \rvert$	Forward transfer admittance	I_D = 2A, V_{DS} = 10V	1.8	3.0	—	S
C_{iss}	Input capacitance	V_{DS} = 25V, V_{GS} = 0V, f = 1MHz	—	600	—	pF
C_{oss}	Output capacitance		—	80	—	pF
C_{rss}	Reverse transfer capacitance		—	12	—	pF
$t_{d\ (on)}$	Turn-on delay time	V_{DD} = 200V, I_D = 2A, V_{GS} = 10V, R_{GEN} = R_{GS} = 50Ω	—	15	—	ns
t_r	Rise time		—	15	—	ns
$t_{d\ (off)}$	Turn-off delay time		—	60	—	ns
t_f	Fall time		—	30	—	ns
V_{SD}	Source-drain voltage	I_S = 2A, V_{GS} = 0V	—	1.5	2.0	V
$R_{th\ (ch-c)}$	Thermal resistance	Channel to case	—	—	1.39	°C/W

PERFORMANCE CURVES

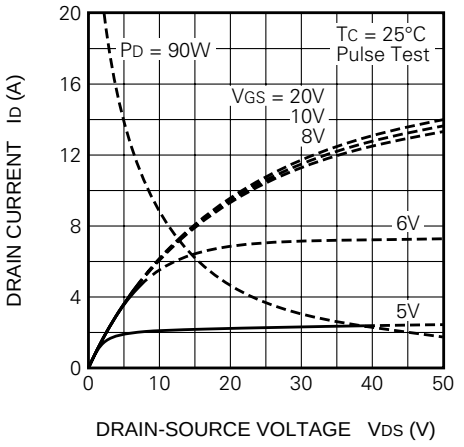
POWER DISSIPATION DERATING CURVE



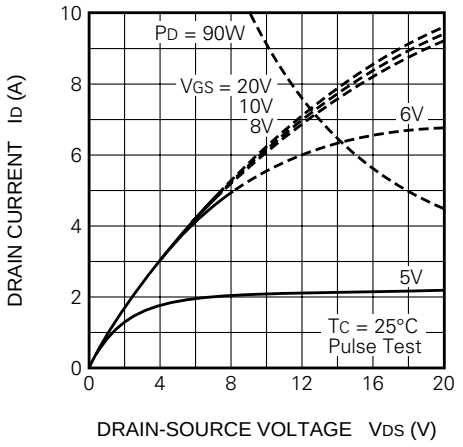
MAXIMUM SAFE OPERATING AREA



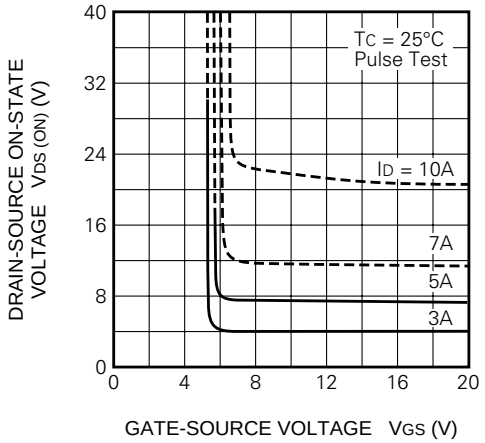
OUTPUT CHARACTERISTICS (TYPICAL)



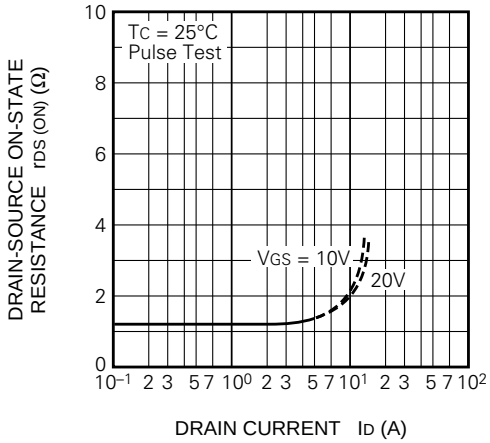
OUTPUT CHARACTERISTICS (TYPICAL)



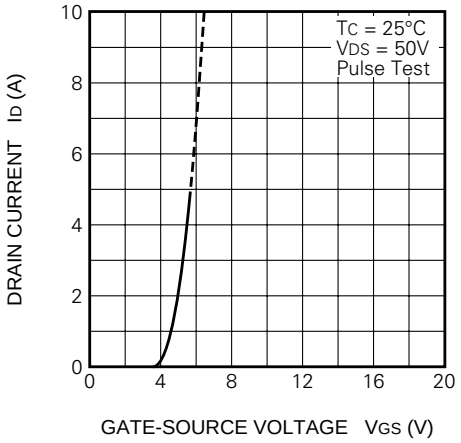
ON-STATE VOLTAGE VS. GATE-SOURCE VOLTAGE (TYPICAL)



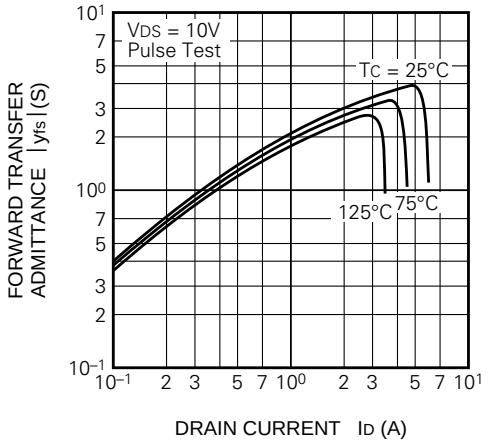
ON-STATE RESISTANCE VS. DRAIN CURRENT (TYPICAL)



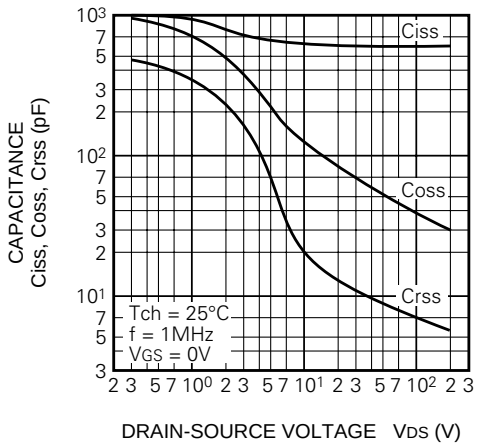
TRANSFER CHARACTERISTICS (TYPICAL)



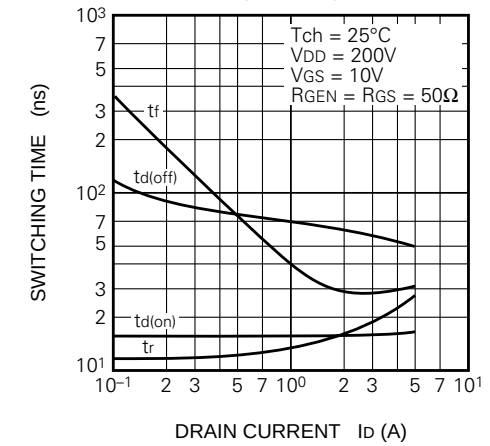
FORWARD TRANSFER ADMITTANCE VS.DRAIN CURRENT (TYPICAL)



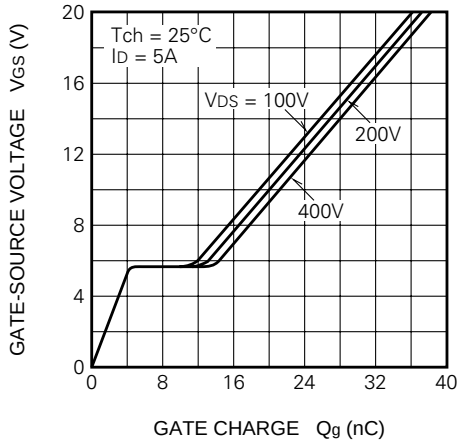
CAPACITANCE VS. DRAIN-SOURCE VOLTAGE (TYPICAL)



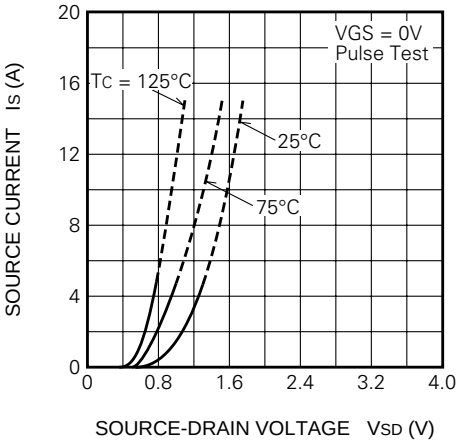
SWITCHING CHARACTERISTICS (TYPICAL)



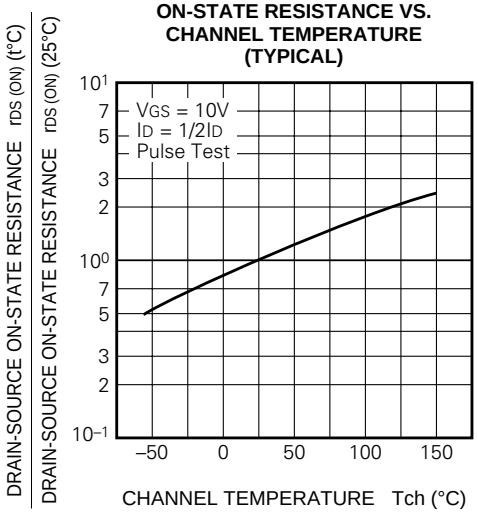
GATE-SOURCE VOLTAGE VS.GATE CHARGE (TYPICAL)



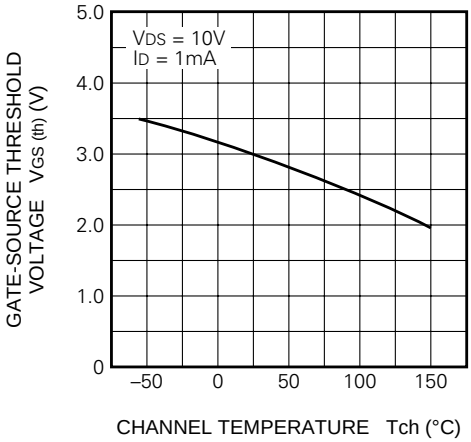
SOURCE-DRAIN DIODE FORWARD CHARACTERISTICS (TYPICAL)



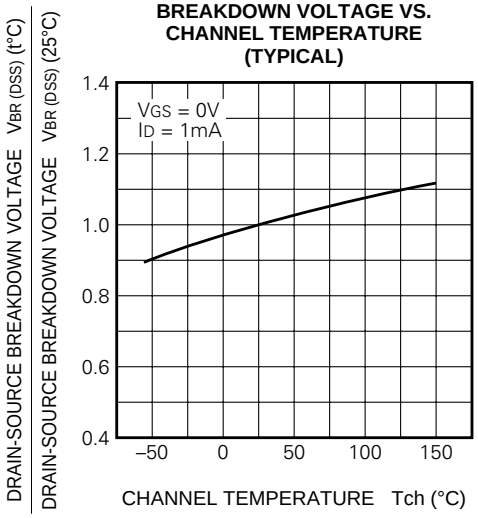
ON-STATE RESISTANCE VS. CHANNEL TEMPERATURE (TYPICAL)



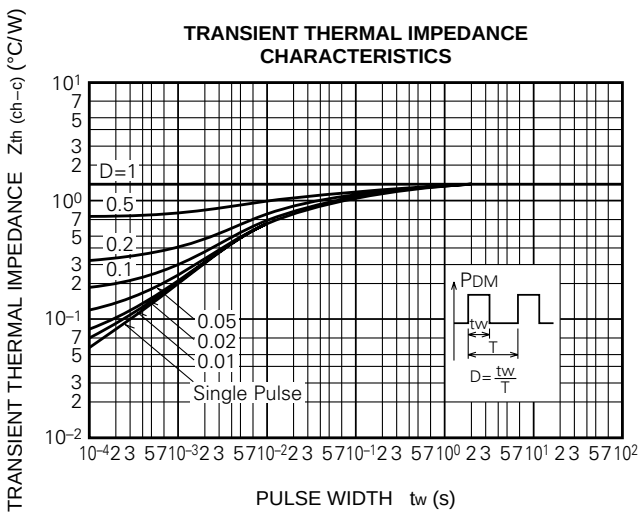
THRESHOLD VOLTAGE VS. CHANNEL TEMPERATURE (TYPICAL)



BREAKDOWN VOLTAGE VS. CHANNEL TEMPERATURE (TYPICAL)



TRANSIENT THERMAL IMPEDANCE CHARACTERISTICS



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